

SKOPJE INTL, North Macedonia

WMO: 135860

Lat: 41.962N

Lon: 21.621E

Elev: 238

StdP: 98.50

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.0	-14.7	1.1	-9.7	-11.5	1.4	-5.9	9.7	5.1	8.4	2.6	0.9	50	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	36.1	20.2	34.3	20.0	32.9	19.7	21.9	32.0	21.2	30.9	20.6	29.9	2.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	24.5	18.1	13.4	23.8	17.5	12.9	23.4	65.0	32.1	62.4	31.2	60.2	30.0	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.3	6.1	-14.4	39.0	4.4	1.8	-17.6	40.3	-20.1	41.4	-22.6	42.4	-25.8	43.7		
(5)			-14.8	23.5	4.2	0.8	-17.9	24.1	-20.4	24.6	-22.7	25.1	-25.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	0.8	3.7	8.0	12.5	17.4	21.7	24.2	19.2	13.5	7.4	2.0	(6)	
(7)		DBStd	8.96	4.50	4.33	3.87	3.38	3.05	3.17	2.71	2.68	3.34	3.80	4.19	4.73	(7)
(8)		HDD10.0	924	285	179	83	15	0	0	0	0	13	98	251		(8)
(9)		HDD18.3	2528	543	410	321	175	54	8	1	1	29	153	327	507	(9)
(10)		CDD10.0	1997	0	2	21	91	229	352	439	440	276	123	21	2	(10)
(11)		CDD18.3	559	0	0	0	1	25	110	182	183	56	4	0	0	(11)
(12)		CDH23.3	6796	0	0	4	49	377	1304	2178	2173	636	75	1	0	(12)
(13)		CDH26.7	3156	0	0	0	8	96	574	1106	1119	238	14	0	0	(13)
(14)	Wind	WSAvg	1.7	1.4	1.8	2.1	1.9	1.6	1.8	2.1	1.6	1.3	1.2	1.3	(14)	
(15)	Precipitation	PrecAvg	519	33	29	36	49	51	47	35	46	56	46	61	(15)	
(16)		PrecMax	748	82	76	74	146	101	94	127	104	147	138	102	311	(16)
(17)		PrecMin	310	2	6	2	15	23	13	0	1	4	0	0	2	(17)
(18)		PrecStd	117	21	20	20	34	20	22	28	24	37	38	27	60	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	18.6	23.3	27.9	31.3	36.2	38.4	38.3	34.6	28.2	21.1	16.0	(19)
(20)			MCWB	9.0	11.4	12.7	16.1	18.4	21.2	20.2	20.2	18.8	17.4	14.4	11.8	(20)
(21)		2%	DB	10.9	15.2	20.7	24.4	29.1	33.9	36.1	36.2	31.9	25.2	18.0	12.1	(21)
(22)			MCWB	7.8	9.6	11.8	14.7	18.0	20.5	20.3	20.1	18.6	16.0	13.5	9.9	(22)
(23)		5%	DB	8.8	12.9	18.2	22.2	27.2	32.0	34.2	34.5	29.4	23.0	16.0	10.1	(23)
(24)			MCWB	6.6	8.5	10.6	13.5	17.4	20.0	20.2	19.8	17.8	15.5	12.6	8.6	(24)
(25)		10%	DB	6.9	10.6	15.9	20.2	25.2	30.1	32.6	32.8	27.1	20.8	14.1	8.2	(25)
(26)			MCWB	5.3	7.3	9.7	12.8	16.7	19.4	19.8	19.5	17.2	15.0	11.3	6.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	12.0	13.7	17.1	20.3	22.8	22.7	22.9	20.7	18.6	15.5	12.5	(27)
(28)			MCD B	12.3	17.3	21.2	25.1	28.0	32.3	32.5	33.1	29.1	25.1	18.7	15.4	(28)
(29)		2%	WB	8.2	10.2	12.4	15.7	19.1	21.7	21.8	21.7	19.6	17.3	14.1	10.2	(29)
(30)			MCD B	10.2	14.2	18.9	22.7	26.7	31.1	32.0	31.9	27.8	22.9	17.2	11.7	(30)
(31)		5%	WB	6.6	8.9	11.2	14.5	18.1	20.8	21.1	21.0	18.8	16.3	12.9	8.9	(31)
(32)			MCD B	8.2	12.4	16.9	20.8	25.6	29.8	31.2	30.6	26.6	21.2	15.7	10.1	(32)
(33)		10%	WB	5.3	7.4	10.0	13.4	17.3	20.0	20.5	20.2	18.0	15.3	11.6	6.9	(33)
(34)			MCD B	6.6	10.1	15.3	18.9	24.0	28.5	30.1	29.7	25.0	19.7	13.8	8.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.0	10.3	12.2	13.3	13.6	14.4	15.0	15.5	13.7	12.5	9.8	7.3	(35)
(36)			MCDBR	10.8	14.9	17.7	17.5	17.3	17.4	18.4	18.8	18.0	16.4	12.4	9.2	(36)
(37)			MCWBR	7.8	9.6	9.8	9.0	7.9	6.6	6.0	6.2	6.8	8.3	7.9	6.8	(37)
(38)		5% WB	MCD BR	9.5	13.5	15.7	15.4	15.6	16.0	16.0	15.6	14.7	13.6	10.7	7.8	(38)
(39)			MCWBR	7.1	9.0	9.1	8.3	7.7	6.5	5.8	5.6	6.1	7.4	7.0	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.360	0.405	0.463	0.456	0.435	0.437	0.454	0.432	0.395	0.362	0.335	(40)	
(41)		taud	2.353	2.271	2.185	2.038	2.128	2.235	2.226	2.174	2.213	2.300	2.347	2.356	(41)	
(42)		Ebn at Noon	799	831	833	808	825	843	836	808	796	782	755	760	(42)	
(43)		Edh at Noon	84	107	131	163	153	138	138	141	126	103	85	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.61	2.50	3.53	4.58	5.58	6.41	6.68	5.99	4.33	3.02	1.92	1.38	(44)	
(45)		RadStd	0.25	0.31	0.43	0.54	0.40	0.50	0.43	0.42	0.36	0.36	0.20	0.24	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.54	N/A	N/A	N/A	N/A	N/A	-150	N/A	N/A

Nomenclature: See separate page